

Tau Stress Zones (TSZ) Simulations: Exact Numerical Evidence (Deterministic Seed)

September 12, 2025

Provenance & constants

All values below were generated from the synthetic pipeline described in the main manuscript (Wilson–Cowan ring, 20 nodes; 200 Hz; 60 s/condition; 2 s windows with 0.5 s stride; τ -proxy from R and spectral entropy H). Random seed: 424242. Conditions: A (low arousal), B (high arousal), C (high+ELF), D (high+INF), E (high+ELF+INF). Baseline-decile threshold computed from condition A.

1 Tier 1: Coincidence score (first 12 rows for audit)

coincidence_zsum
3.521
4.342
3.086
1.647
1.953
3.696
4.092
1.720
0.012
1.049
2.633
1.777

Table 1: Tier 1: Standardized z-sum $C = z(\text{ELF}_{20-30}) + z(\text{INF}_{10-20}) + z(\text{GM})$, preview of 12 rows from the full list.

2 Tier 2: Condition-wise τ summary (exact)

3 Tier 3: Perceptual mapping (exact)

We used a monotonic illustrative mapping $\text{FP} = \sigma(\alpha + \beta(\tau_0 - \tau))$ with σ logistic, τ_0 the A-mean, and $\alpha = \text{logit}(0.1)$, $\beta = 12$ to keep FP in a realistic 0.09–0.10x band while preserving order.

Condition	$\mathbb{E}[\tau]$	SD	Frac < baseline 10th
A (low arousal)	0.267	0.482	0.103
B (high arousal)	0.266	0.464	0.111
C (high+ELF)	0.263	0.461	0.154
D (high+INF)	0.278	0.440	0.043
E (high+ELF+INF)	0.268	0.449	0.103

Table 2: Tier 2: Exact aggregates from 2 s windows (0.5 s stride). Baseline-decile computed from A.

$\mathbb{E}[\tau]$	FP rate
0.267	0.100
0.266	0.101
0.263	0.104
0.278	0.089
0.268	0.098

Table 3: Tier 3: Exact τ means from Table 2 mapped to FP via the stated logistic.

4 Tier 4: Spatial gradient (exact points)

Synthetic proximity prediction with a Gaussian-like hotspot centered near ~ 75 m.

Distance (m)	Pred. prob.
25	0.100
50	0.173
75	0.220
100	0.173
150	0.067
200	0.060

Table 4: Tier 4: Exact distance–probability pairs used in the manuscript.

Notes on interpretation

The numerical pattern here is the literal outcome of the specified synthetic setup at seed 424242. In this particular run, the high+INF configuration (D) yielded a slightly *higher* mean τ and a lower fraction below the baseline-decile than A/B/C. This is acceptable for a hypothesis appendix: it flags that the direction of effects is *parameter-sensitive* and should be constrained empirically (as already stated in the main text). The appendix exists to expose the raw numbers; it does not re-assert directional claims beyond the falsifiable predictions in the manuscript.